

Homework-3

1. Let k be a field, $k(x)$ the function field in one variable, i.e. quotient field of $k[x]$. Let $f(x), g(x) \in k[x]$, $g(x) \neq 0$. Is the field extension $k(x)$ over $k\left(\frac{f(x)}{g(x)}\right)$ algebraic? Transcendental? finite? If finite, what is $[k(x) : k\left(\frac{f(x)}{g(x)}\right)]$?
2. Let $\alpha_1, \dots, \alpha_n$ be algebraic over k . We proved $[k(\alpha_1, \dots, \alpha_n) : k] \leq \prod_{i=1}^n [k(\alpha_i) : k]$. Give an example where the inequality is strict.
3. Prove that (i) $\mathbb{Q}(2^{1/3} + \omega) = \mathbb{Q}(2^{1/3}, \omega)$. Here $\omega = e^{2\pi i/3}$.
(ii) $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Find the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$.
4. Show that $\mathbb{Q}(\sqrt{5} + \sqrt{7}) = \mathbb{Q}(\sqrt{5}, \sqrt{7})$.
5. (Universal property for polynomial rings) Let A be a ring containing a field F . Let $a_1, \dots, a_n \in A$. Show that $\exists!$ ring homomorphism $\varphi: F[x_1, \dots, x_n] \rightarrow A$ with $\varphi(x_i) = a_i, 1 \leq i \leq n$.